

Homework — 2.2.1 Programming Techniques

Name: _____ Date: _ **Mark:** ____ / 20

OCR H446 — set after the 2.2.1 lesson. Show traces/working. Code uses OCR Exam Reference Language.

Part A — This week's topic (~14 marks)

1. State the difference between **count-controlled** and **condition-controlled** iteration, and name **one** OCR Exam Reference Language construct used for each. **[2]** (AO1)

2. A program validates input by asking again for a value until the value entered is greater than 0. Explain why a **do ... until** loop is more suitable here than a **while** loop. **[2]** (AO1)

3. Trace the call **power(2, 4)** for the recursive function below. Show the value returned at **each** level of the recursion and the final result. **[3]** (AO2)

```
function power(base, exp)
  if exp == 0 then
    return 1
  else
    return base * power(base, exp - 1)
  endif
endfunction
```

4. Study the program below. In OCR Exam Reference Language, a variable assigned inside a subroutine is **local** unless it is declared **global**.

```
count = 8

procedure reset()
    count = 0
    print(count)
endprocedure

reset()
print(count)
```

(a) State the two values printed, in order. **[1]** (AO2)

Output line 1: _ *Output line 2:* _

(b) Explain your answer to part (a), referring to the **scope** of the variables. **[2]** (AO2)

5. The two procedures below are each called with the variable **total**, which holds the value 10.

```
procedure scaleByVal(n:byVal)
    n = n * 2
endprocedure

procedure scaleByRef(n:byRef)
    n = n * 2
endprocedure
```

State the value of **total** after each call below and justify each answer. **[2]** (AO2)

After `scaleByVal(total)` : total = ___ **because** ____

After `scaleByRef(total)` : total = ___ **because** ____

6. Define the term **encapsulation** and state **one** benefit it provides in object-oriented programming. [2]
(AO1)

Part B — Retrieval: keep it fresh (~6 marks)

7. (2.1.1) State what is meant by **abstraction** and give **one** reason it is useful when solving a problem. [2]

8. (1.4.2) A **stack** and a **queue** are both data structures. State the order in which items are removed from each (use the correct three-letter acronym). [2]

- Stack: _____
- Queue: _____

9. (2.1.3) Explain what is meant by **decomposition**. [2]
